

Soil pollution

- Soil pollution is defined as the introduction of substances, biological organisms or energy into the soil, resulting in a change in the soil quality, likely to effect the normal use of the soil or endangering public health and the living environment.
- Sources of soil pollution
 - Urban wastes
 - Industrial waste
 - Agricultural practices
 - Soil conditioners
 - Farm house wastes
 - Radioactive wastes
 - Biological agents



- Urban wastes
 - Domestic and commercial solid wastes, termed as 'refuse' containing rubbish materials like papers, fibers, plastics, glasses, bottles, street sweepings, leaves, abandoned vehicles and other discarded products
- Industrial wastes
 - Consisting of inorganic and organic compounds , majority being non-biodegradable materials in solid and liquid wastes dumped on soil. Accumulation of these substances are triggered by the role of the soil as filtering medium.

Industry	Contaminants
Paper Mill	Chloride, Sodium
Sugar	Nitrogen, Phosphorus
Steel and Coke	Cyanide, phenols
Refineries	Phosphate, sulphur, chromium, phenol
Fertilizers	Chloride, Nitrogen, Phosphorus, Potassium
Pesticides	Organic nitrogen, sulphate

- Agricultural practices
 - To save the crops and their products from insects, fungi, bacteria, virus, rodents, in modern agricultural practices, insecticides, fungicides and herbicides etc are used, which accumulate in soil. Some pests develop resistance, so more or higher dose is required, which also accumulate in soil. Chemicals such as Aldrin, endrin, dieldrin, endosulphan also enters food chain.
- Soil conditioners
 - Soil conditioners are used to increase the fertility of soil. Generally they contain toxic metals such as Arsenic, Cadmium, Pb, Hg.
- Farm house waste
 - The faecal matter of animal husbandry and poultry enriches the soil with N and P, which causes undesirable effects on soil texture

- Radioactive wastes
 - Generally storage and disposal of radioactive wastes causes penetration and accumulation of radioactive materials. The radio nucleids such as isotopes of Sr-90, I-129 and 131, Caesium 137, Ruthenium 106, Barium 140, Caesium 144, lanthanum 140 deposits in the top soil and emits gamma radiation
- Biological agents
 - Human, animal and bird excreta. The pathogenic organisms that pollute the soil are classified under three categories
 - a. Pathogens originating in soil
 - b. Pathogens excreted by humans
 - c. Pathogens excreted by animals



Effects of Soil Pollution

- Removal of top soil causes low fertility for crop production
- Accumulation of toxic chemicals
- Soil natural flora and fauna beneficial for environment are adversely affected
- Bioaccumulation in crops
- Lead to leaching and groundwater pollution
- Metal induced toxicities in human beings

Control of soil pollution

- Controlling erosion and preserving the top soil
- Disposal of properly treated industrial wastes
- Preventing the entry of leachates from landfill sites into the soil beneath
- Use of correct measure of fertilizers and pesticides
- Crop rotation can increase the fertility of soil
- More plough and mix improve aeration, porosity and permeability of soil

Biomedical wastes

- **Biomedical waste** or **hospital waste** is any kind of waste containing infectious (or potentially infectious) materials.
- It may also include waste associated with the generation of biomedical waste that visually appears to be of medical or laboratory origin (e.g. packaging, unused bandages, infusion kits etc.), as well research laboratory waste containing biomolecules or organisms that are mainly restricted from environmental release.



Types

- Biomedical waste may be solid or liquid. Examples of infectious waste include
 - discarded blood,
 - sharps,
 - unwanted microbiological cultures and stocks,
 - identifiable body parts (including those as a result of amputation),
 - other human or animal tissue,
 - used bandages and dressings, discarded gloves, other medical supplies that may have been in contact with blood and body fluids, and laboratory waste that exhibits the characteristics described above.
 - Waste sharps include potentially contaminated used (and unused discarded) needles, scalpels, lancets and other devices capable of penetrating skin.

Effects on humans

- Disposal of this waste is an environmental concern, as many medical wastes are classified as *infectious* or *biohazardous* and could potentially lead to the spread of infectious disease.
- The most common danger for humans is the infection which also affects other living organisms in the region.
- Daily exposure to the wastes (landfills) leads to accumulation of harmful substances or microbes in the person's body.

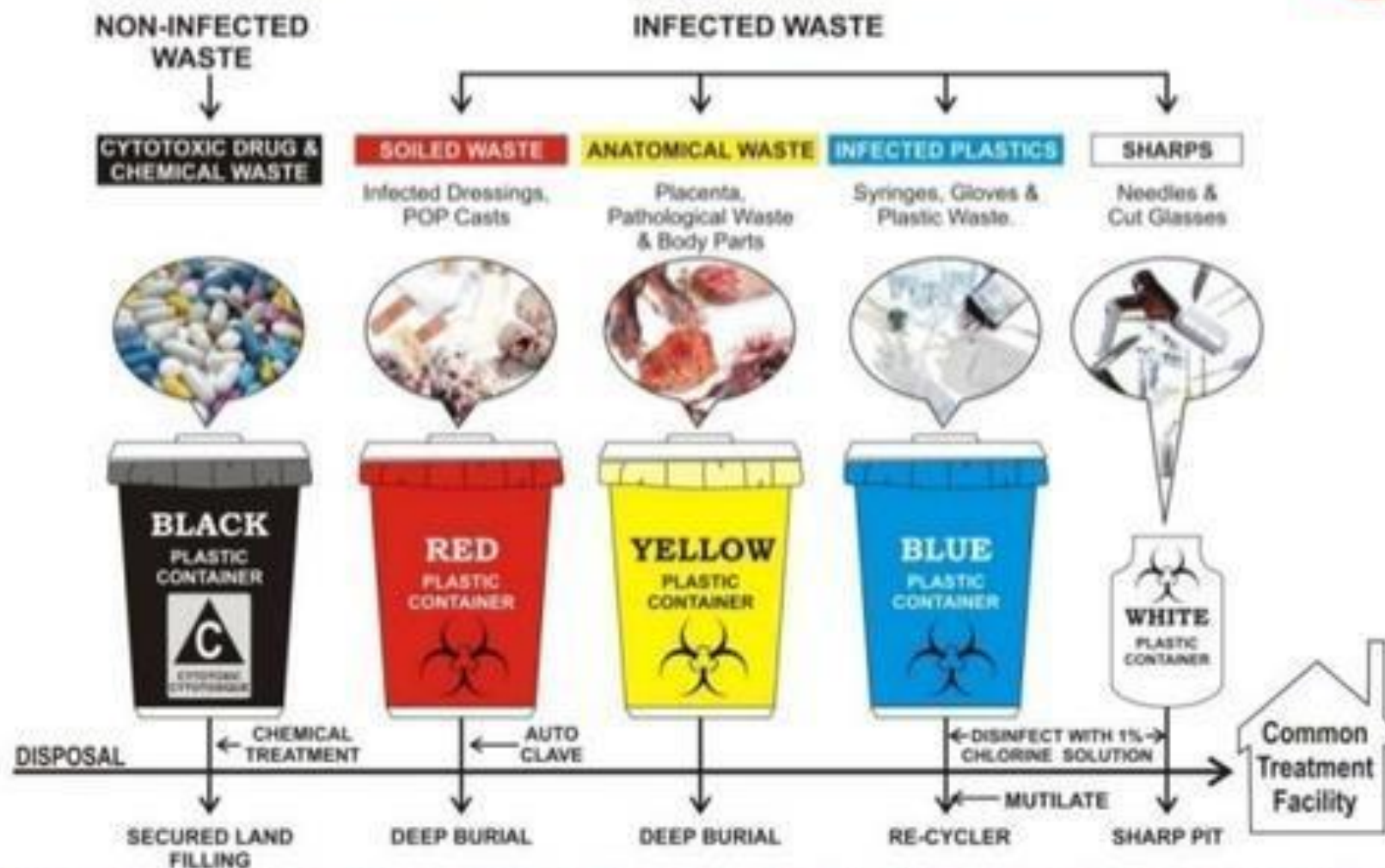
Management

- Biomedical waste must be properly managed and disposed of to protect the environment, general public and workers, especially healthcare and sanitation workers who are at risk of exposure to biomedical waste as an occupational hazard.
- Steps in the management of biomedical waste include generation, accumulation, handling, storage, treatment, transport and disposal
 - **Generation and accumulation** - Biomedical waste should be collected in containers that are leak-proof and sufficiently strong to prevent breakage during handling. Containers of biomedical waste are marked with a biohazard symbol.
 - Discarded sharps are usually collected in specialized boxes, often called *needle boxes*.





SEGREGATION OF SOLID BIO-MEDICAL WASTE



NOTE:- USE ANY COLORED BIN OTHER THAN BLACK, RED, YELLOW, BLUE & WHITE FOR DISPOSAL OF GENERAL WASTE

- **Storage & Handling**

- Storage refers to keeping the waste until it is treated on-site or transported off-site for treatment or disposal. There are many options and containers for storage. Regulatory agencies may limit the time for which waste can remain in storage.
- Handling is the act of moving biomedical waste between the point of generation, accumulation areas, storage locations and on-site treatment facilities. Workers who handle biomedical waste must observe *standard precautions*

Treatment

- Biomedical waste is often incinerated. An efficient incinerator will destroy pathogens and sharps. Source materials are not recognizable in the resulting ash. Alternative thermal treatment can also include technologies such as gasification and pyrolysis including energy recovery with similar waste volume reductions and pathogen destruction.
- An autoclave may also be used to treat biomedical waste. An autoclave uses steam and pressure to sterilize the waste or reduce its microbiological load to a level at which it may be safely disposed of.
- Microwave disinfection can also be employed for treatment of Biomedical wastes. Microwave irradiation is a type of non-contact heating technologies for disinfection.
- For liquids and small quantities, a 1–10% solution of bleach can be used to disinfect biomedical waste. Solutions of sodium hydroxide and other chemical disinfectants may also be used, depending on the waste's characteristics.
- For autoclaves and microwave systems, a shredder may be used as a final treatment step to render the waste unrecognizable. Some autoclaves have built in shredders.